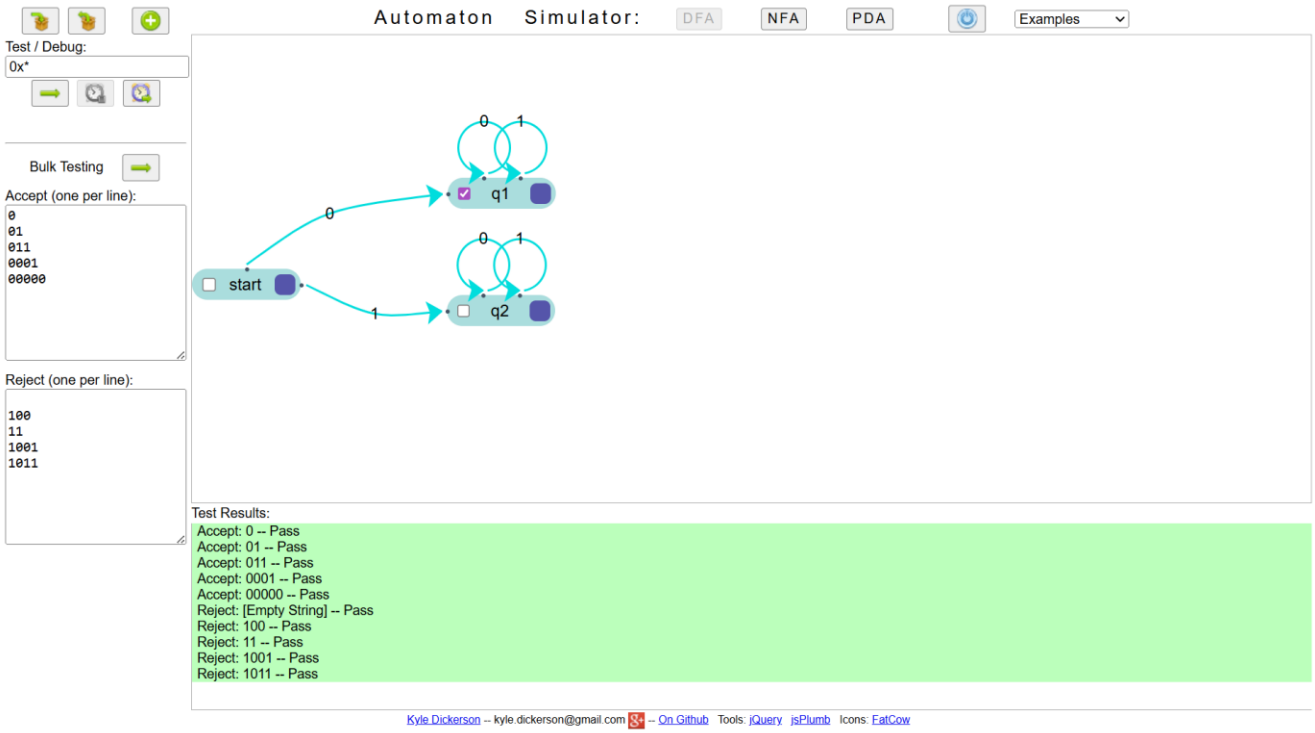
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{0, 1\}$, que acepte el conjunto de palabras del lenguaje $\boldsymbol{L = \{0x \mid x \in \{0,1\}*\}}$

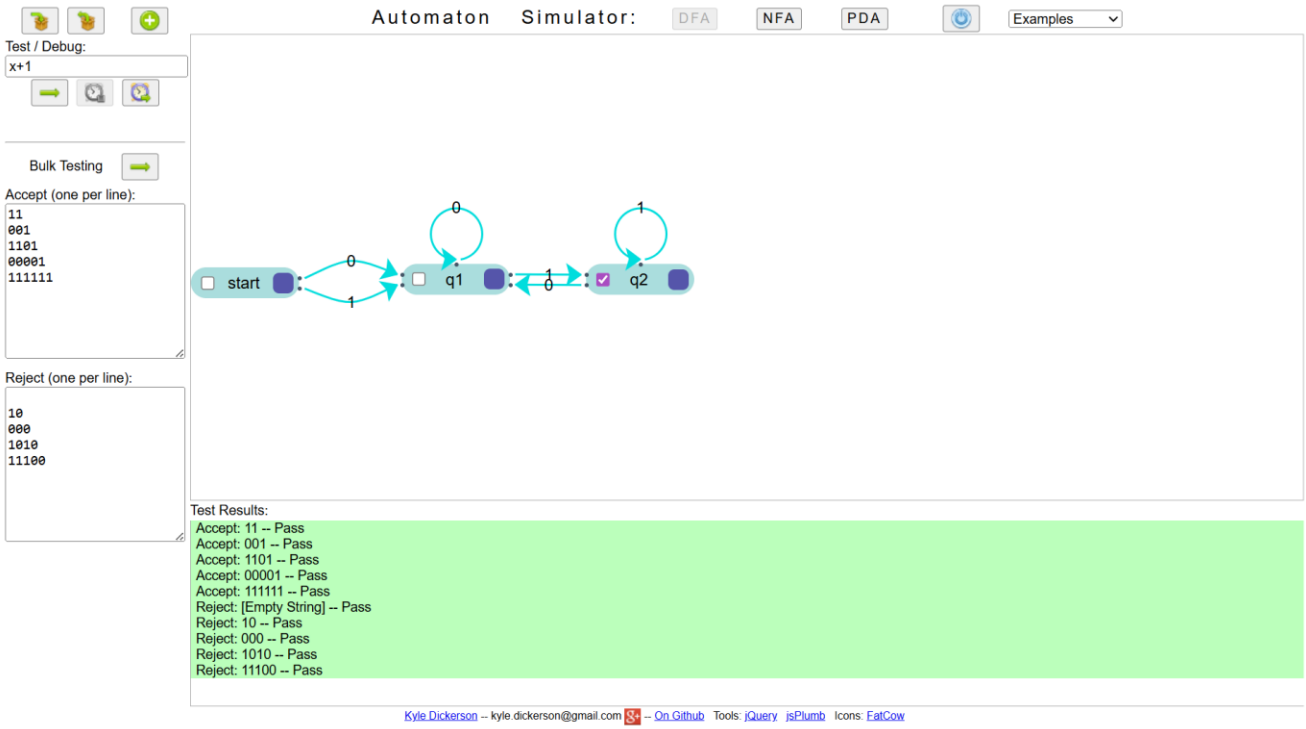



$$\boldsymbol{L = 0x | x \in \{0,1\}^*\}}$$

| ***Elementos*** | **Valores** |
|---|---|
| *Σ* | {0,1} |
| *Q* | {q0, q1, q2} |
| *δ* | f(q0,0) = q1<br>f(q0,1) = q2<br>f(q1,0) = q1<br>f(q1,1) = q1<br>f(q2,0) = q2<br>f(q2,1) = q2 |
| *q0* | q0 |
| *F* | {q1} |

| ***Estado*** | **0** | **1** |
|---|---|---|
| → *q0* | q1 | q2 |
| * *q1* | q1 | q1 |
| *q2* | q2 | q2 |

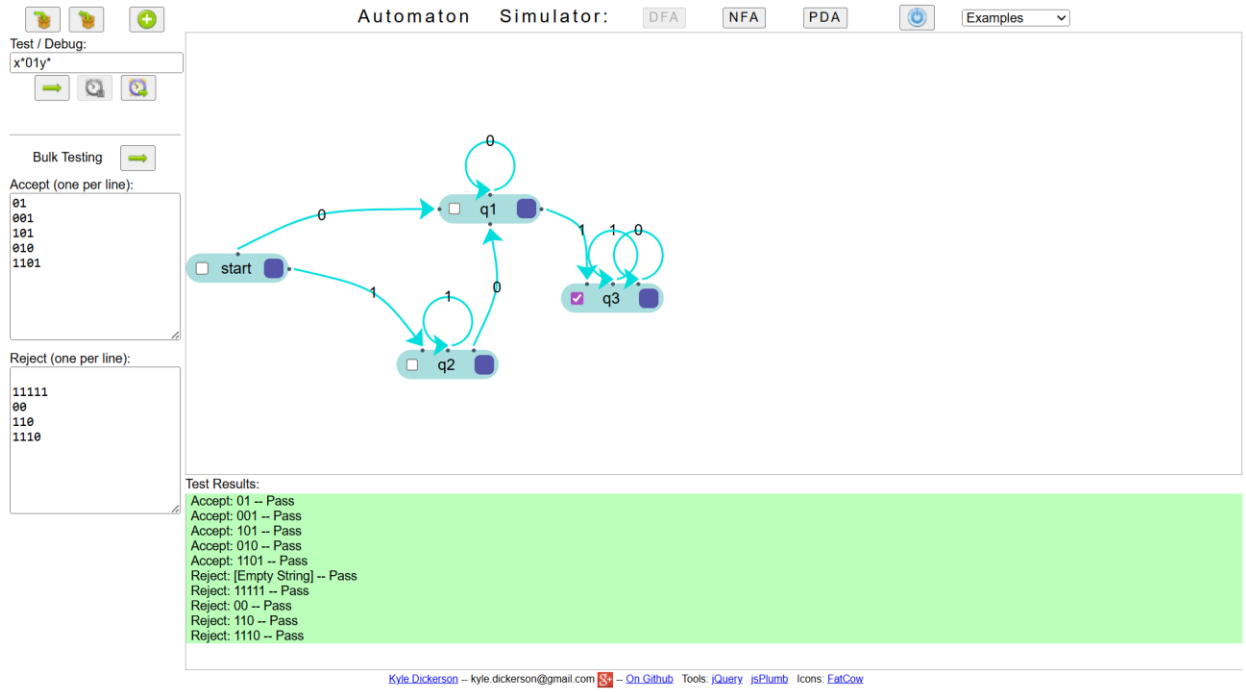
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{0, 1\}$, que acepte el conjunto de palabras del lenguaje $\boldsymbol{L = \{x1 \mid x \in \{0,1\})\ \}}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {0,1} |
| $Q$ | {q0, q1, q2} |
| $\delta$ | f(q0,0) = q1<br>f(q0,1) = q1<br>f(q1,0) = q1<br>f(q1,1) = q2<br>f(q2,0) = q1<br>f(q2,1) = q2 |
| *q0* | q0 |
| $F$ | {q2} |

| ***Estado*** | **0** | **1** |
|---|---|---|
| → *q0* | q1 | q1 |
| **q1* | q1 | q2 |
| *q2* | q1 | q2 |

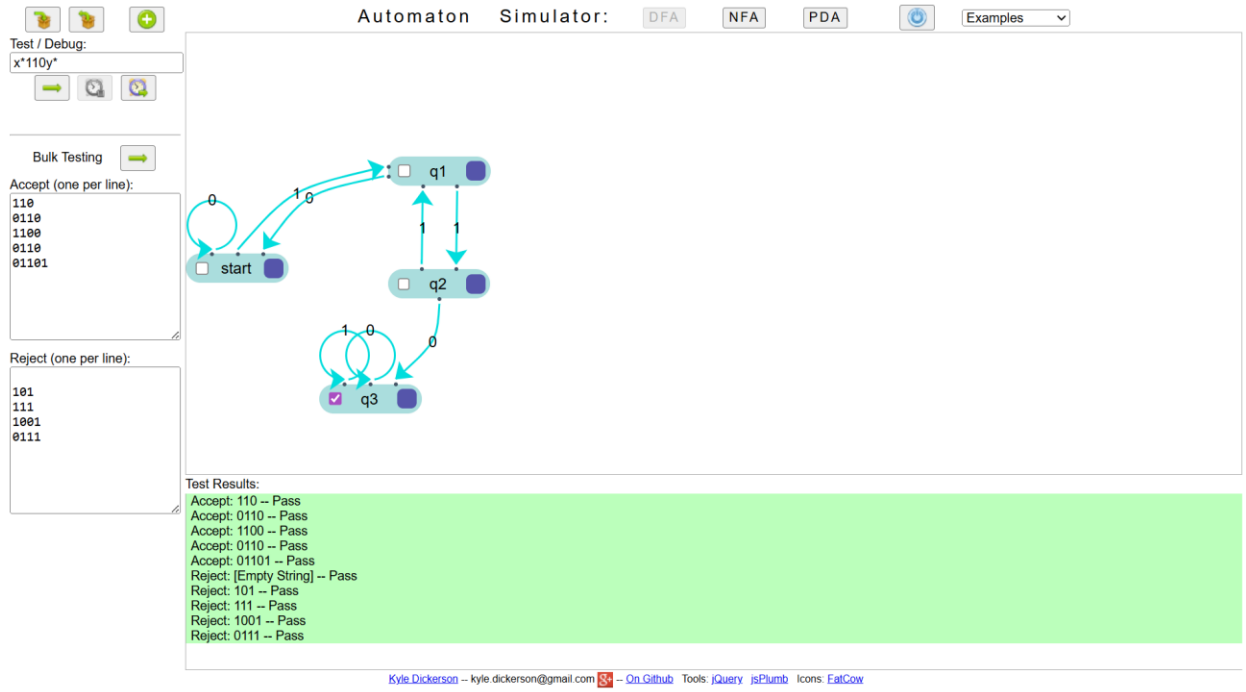
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{0, 1\}$, que acepte el conjunto de palabras del lenguaje $\boldsymbol{L = \{x01y \mid x, y \in \{0,1\}*\}}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {0,1} |
| $Q$ | {q0, q1, q2, q3} |
| $\delta$ | f(q0,0) = q1<br>f(q0,1) = q2<br>f(q1,0) = q1<br>f(q1,1) = q3<br>f(q2,0) = q1<br>f(q2,1) = q2<br>f(q3,0) = q3<br>f(q3,1) = q3 |
| *q0* | q0 |
| $F$ | {q3} |

| ***Estado*** | **0** | **1** |
|---|---|---|
| → *q0* | q1 | q2 |
| *q1* | q1 | q3 |
| *q2* | q1 | q2 |
| * *q3* | q3 | q3 |

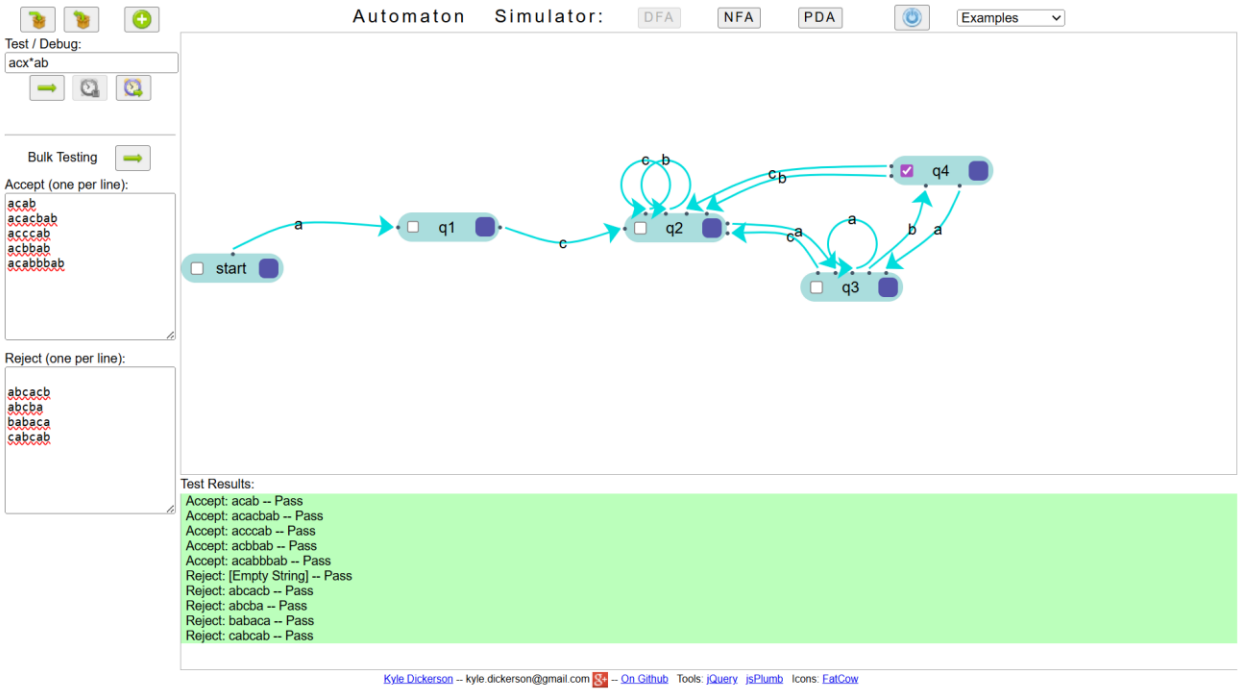
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{0, 1\}$, que acepte el conjunto de palabras del lenguaje $\boldsymbol{L = \{x110y \mid x, y \in \{0,1\}*\}}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {0, 1} |
| $Q$ | {q0, q1, q2, q3} |
| $\delta$ | f(q0, 0)=q0<br>f(q0, 1)=q1<br>f(q1, 0)=q0<br>f(q1, 1)=q2<br>f(q2, 0)=q3<br>f(q2, 1)=q1<br>f(q3, 0)=q3<br>f(q3, 1)=q3 |
| *q0* | q0 |
| $F$ | {q3} |

| ***Estado*** | **0** | **1** |
|---|---|---|
| → *q0* | q0 | q1 |
| *q1* | q0 | q2 |
| *q2* | q3 | q1 |
| * *q3* | q3 | q3 |

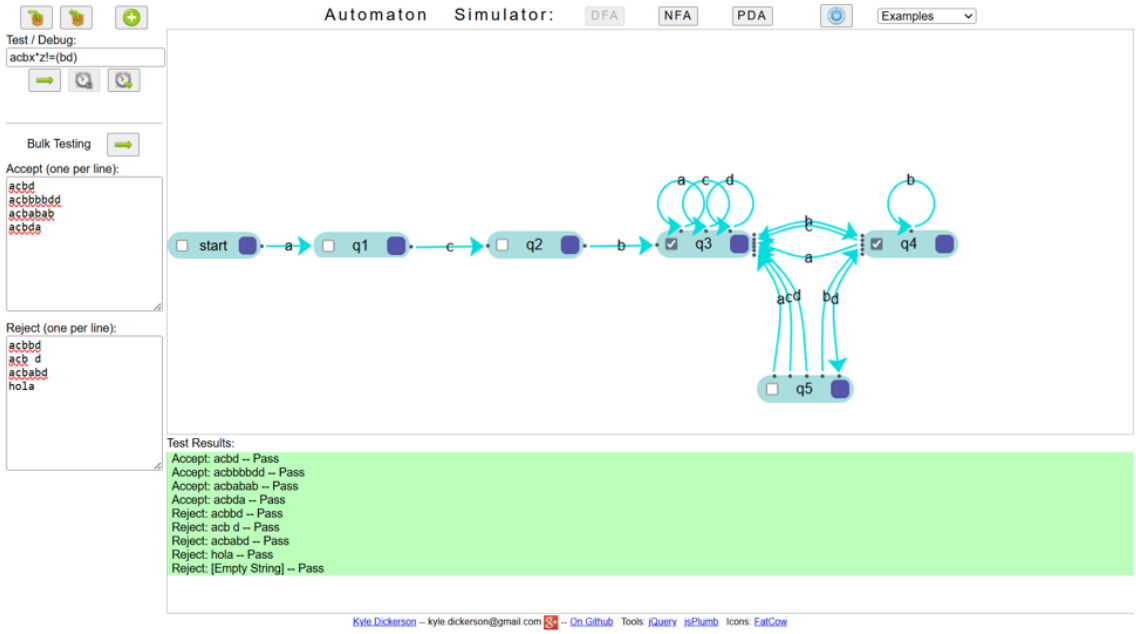
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{a, b, c\}$, que acepte el conjunto de palabras del lenguaje $L = \{acxab \mid x \in \{a, b, c\}*\}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {a, b, c} |
| $Q$ | {q0, q1, q2, q3, q4} |
| $\delta$ | f(q0, a) = q1<br>f(q1, c) = q2<br>f(q2, b) = q2<br>f(q2, c) = q2<br>f(q2, a) = q3<br>f(q3, a) = q3<br>f(q3, b) = q4<br>f(q3, c) = q2<br>f(q4, b) = q2<br>f(q4, c) = q2<br>f(q4, a) = q3 |
| *q0* | q0 |
| $F$ | {q4} |

| ***Estado*** | **a** | **b** | c |
|---|---|---|---|
| → *q0* | q1 | | |
| *q1* | | | q2 |
| *q2* | q3 | q2 | q2 |
| *q3* | q3 | q4 | q2 |
| **q4* | q3 | q2 | q2 |

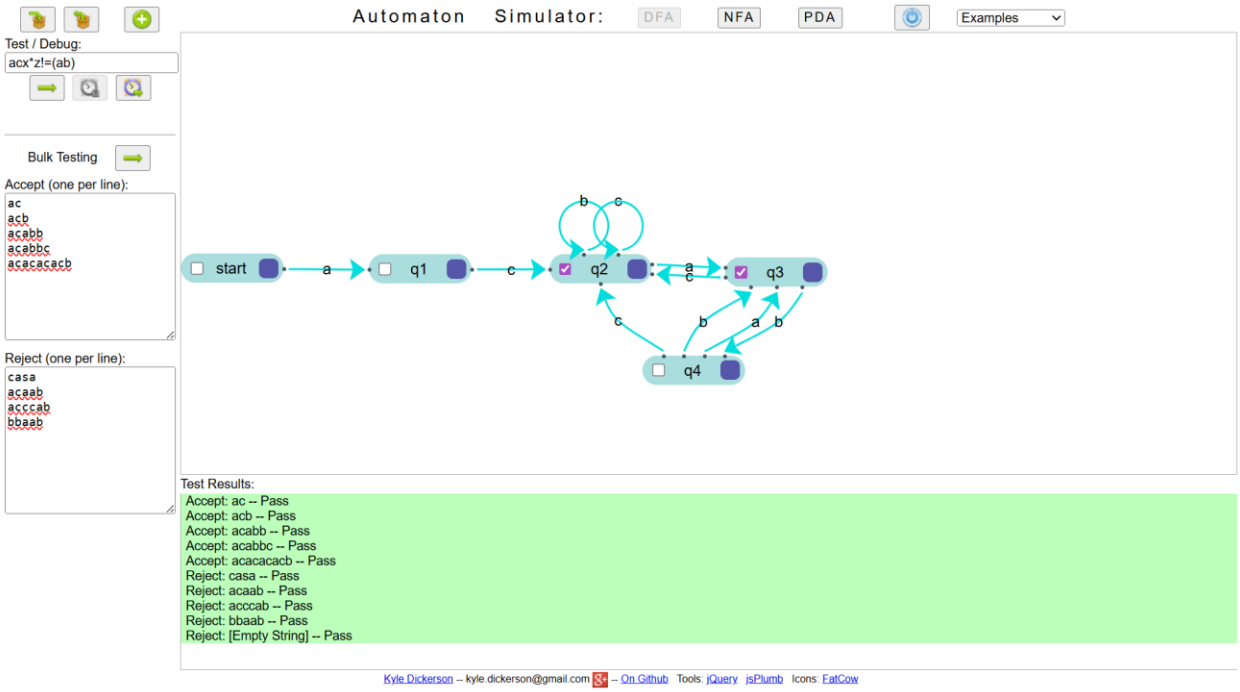
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{a, b, c, d\}$, que acepte el conjunto de palabras del lenguaje $\boldsymbol{L = \{acbxz \mid x \in \{a, b, c, d\}* \; y \; z \neq bd\}}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {a, b, c, d} |
| $Q$ | {q0, q1, q2, q3, q4, q5} |
| $\delta$ | f(q0, a)=q1<br>f(q1, c)=q2<br>f(q2, b)=q3<br>f(q3, a)=q3<br>f(q3, c)=q3<br>f(q3, d)=q3<br>f(q4, a)=q3<br>f(q4, b)=q4<br>f(q4, d)=q5<br>f(q5, a)=q3<br>f(q5, c)=q3<br>f(q5, d)=q3 |
| *q0* | q0 |
| *F* | {q3, q4} |

| ***Estado*** | **a** | **b** | c | d |
|---|---|---|---|---|
| → *q0* | q1 | | | |
| *q1* | | | q2 | |
| *q2* | | q2 | | |
| **q3* | q3 | q4 | q3 | q3 |
| **q4* | q3 | q4 | q3 | q5 |
| q5 | q3 | q4 | q3 | q3 |

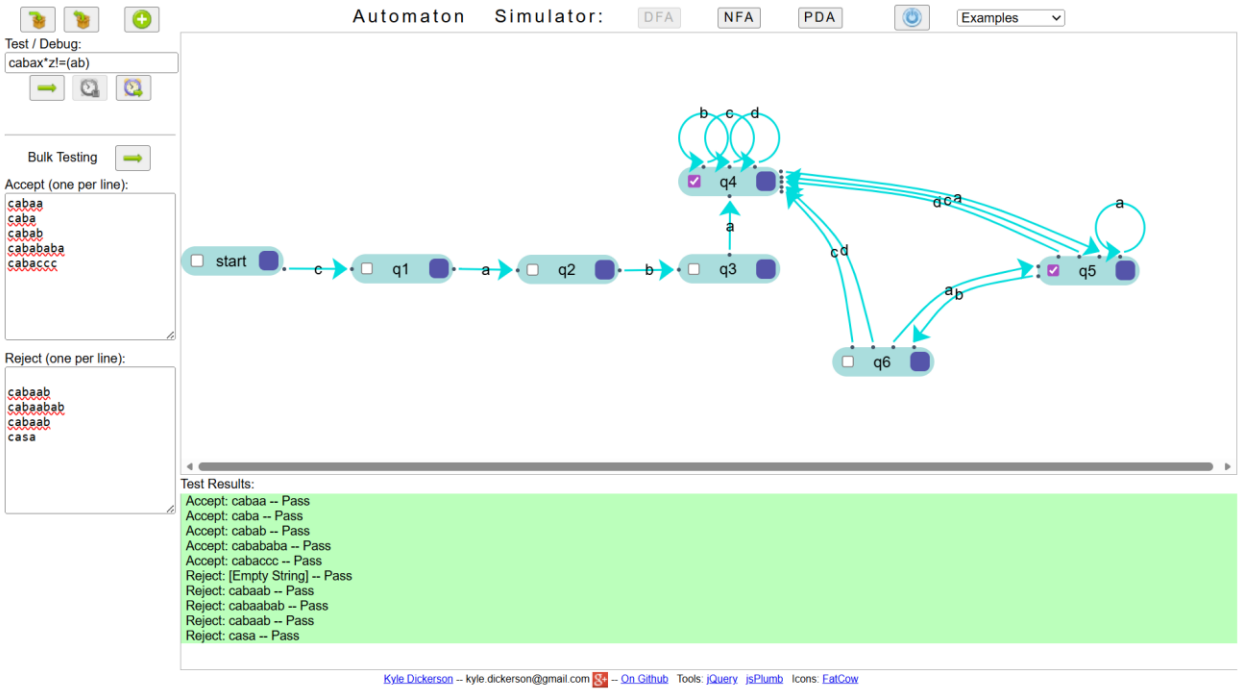
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{a, b, c\}$, que acepte el conjunto de palabras del lenguaje $L = \{acxz \mid x \in \{a. b, c\}* \; y \; z \neq ab \}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {a, b, c} |
| $Q$ | {q0, q1, q2, q3, q4,} |
| $\delta$ | f(q0, a)=q1<br>f(q1, c)=q2<br>f(q2, b)=q2<br>f(q2, c)=q2<br>f(q2, a)=q3<br>f(q3, c)=q2<br>f(q3, a)=q4<br>f(q3, b)=q4<br>f(q4, c)=q2<br>f(q4, a)=q3<br>f(q4, b)=q3 |
| *q0* | q0 |
| *F* | {q2, q3} |

| ***Estado*** | **a** | **b** | **c** |
|---|---|---|---|
| → *q0* | q1 | | |
| *q1* | | | q2 |
| **q2* | q3 | q2 | q2 |
| **q3* | q4 | q4 | q2 |
| *q4* | q3 | q3 | q2 |

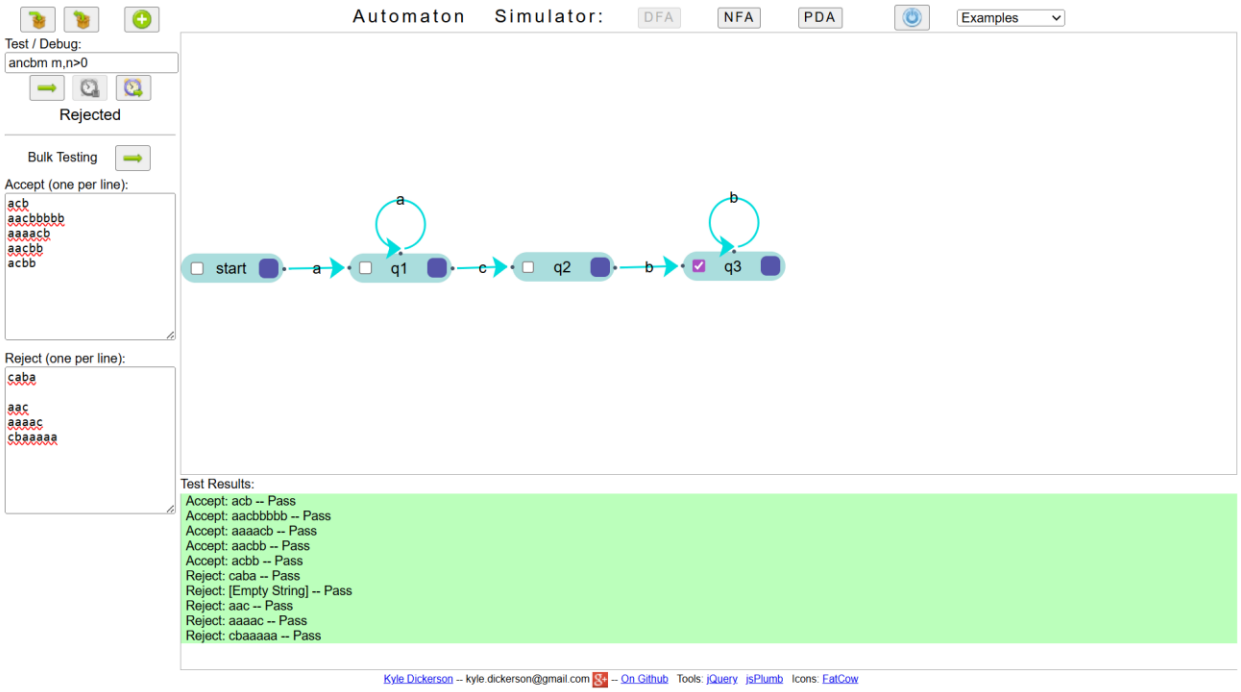
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{a, b, c, d\}$, que acepte el conjunto de palabras $\boldsymbol{L = \{cabaxz \mid x \in \{a, b, c, d\}* \ y \ z \neq ab\}}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {a, b, c} |
| $Q$ | {q0, q1, q2, q3, q4, q5 , q6} |
| $\delta$ | f(q0,c)=q1<br>f(q1,a)=q2<br>f(q2,b)=q3<br>f(q3,a)=q4<br>f(q4,a)=q5<br>f(q4,b)=q4<br>f(q4,c)=q4<br>f(q4,d)=q4<br>f(q5,a)=q5<br>f(q5,b)=q6<br>f(q5,c)=q4<br>f(q5,d)=q4<br>f(q6,a)=q5<br>f(q6,b)=q4<br>f(q6,c)=q4<br>f(q6,d)=q4 |
| *q0* | q0 |
| *F* | {q4,q5} |

| ***Estado*** | **a** | **b** | **c** | **d** |
|---|---|---|---|---|
| → *q0* | | | q1 | |
| *q1* | q2 | | | |
| *q2* | | q3 | | |
| *q3* | q4 | | | |
| **q4* | q5 | q4 | q4 | q4 |
| **q5* | q5 | q6 | q4 | q4 |
| *q6* | q5 | q4 | q4 | q4 |

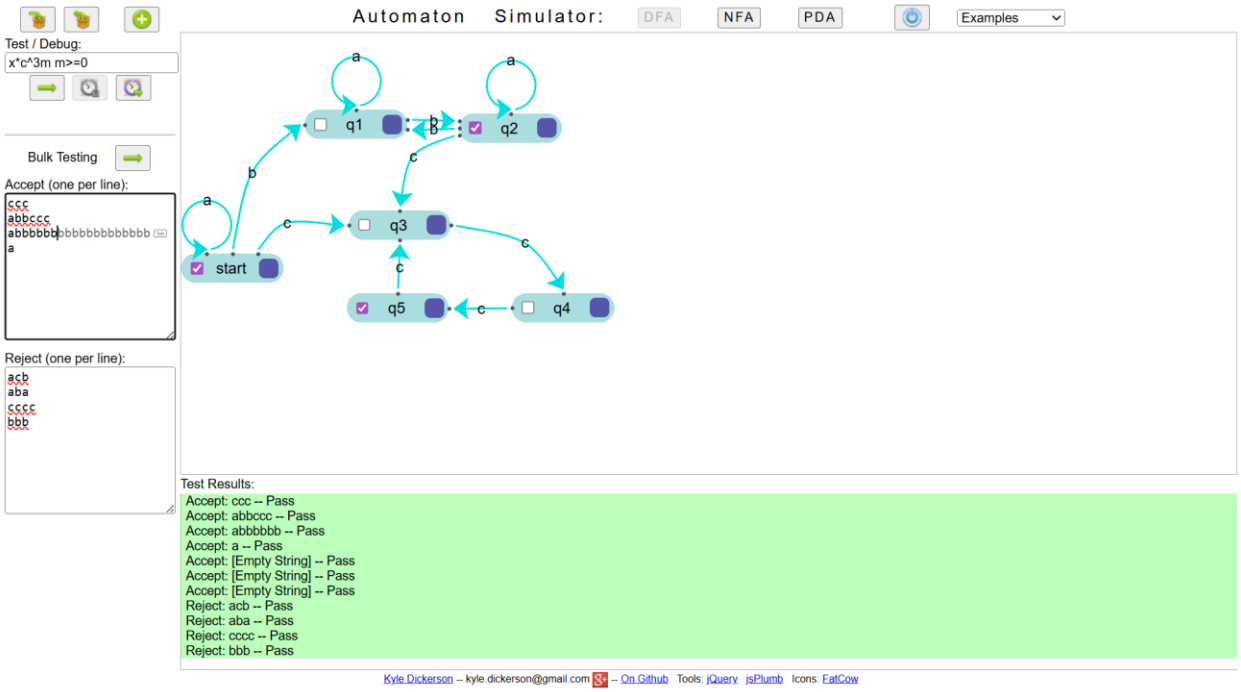
Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{a, b, c\}$, que acepte el lenguaje $\boldsymbol{L = \{a^n c b^m \mid n > 0 \ y \ m > 0\}}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {a, b, c} |
| $Q$ | {q0, q1, q2, q3} |
| $\delta$ | f(q0, a) =q1<br>f(q1, a) =q1<br>f(q1, c) =q2<br>f(q2, b) =q3<br>f(q3, b) =q3 |
| *q0* | q0 |
| $F$ | {q3} |

| ***Estado*** | **a** | **b** | c |
|---|---|---|---|
| → *q0* | q1 | | |
| *q1* | q1 | | q2 |
| *q2* | q2 | q3 | |
| **q3* | | q3 | |

Obtenga un Autómata Finito Determinista (AFD) dado el lenguaje definido en el alfabeto $\Sigma = \{a, b, c\}$, que acepte el lenguaje $\boldsymbol{L = \{xc{+}{*} \mid x \in \{a, b\}* \text{ y la cantidad de } b, s \text{ es par y } m \geq 0\}}$




| ***Elementos*** | **Valores** |
|---|---|
| $\Sigma$ | {a, b, c} |
| $Q$ | {q0, q1, q2, q3, q4, q5} |
| $\delta$ | f(q0, a)=q0<br>f(q0, b)=q1<br>f(q0, c)=q3<br>f(q1, a)=q1<br>f(q1, b)=q2<br>f(q2, a)=q2<br>f(q2, c)=q3<br>f(q3, c)=q4<br>f(q4, c)=q5<br>f(q5, c)=q3 |
| $q0$ | q0 |
| $F$ | {q0, q2, q5} |

| ***Estado*** | **a** | **b** | **c** |
|---|---|---|---|
| $\rightarrow q0$ | q0 | q1 | q3 |
| $q1$ | q1 | q2 | |
| $*q2$ | q2 | | q3 |
| $q3$ | | | q4 |
| $q4$ | | | q5 |
| $*q5$ | | | q3 |